

Independent Replication of Dürr & Høyer (1996):
“A Quantum Algorithm for Finding the Minimum”
arXiv:quant-ph/9607014

Independent replicator (subagent)

2026-07-06

1 Paper summary

Let $T[0..N-1]$ be an unsorted table of N distinct comparable values. The minimum-searching problem is to find the index y^* such that $T[y^*] = \min_j T[j]$. Classically this takes $\Theta(N)$ probes on any deterministic or randomized Turing machine, and a linear-lower-bound argument matches it. Dürr & Høyer¹ present a quantum algorithm running in $O(\sqrt{N})$ Grover queries, matching the known $\Omega(\sqrt{N})$ lower bound up to constant factor.

The algorithm:

1. Draw an initial threshold index y uniformly from $\{0, \dots, N-1\}$.
2. Repeat until total running time exceeds $22.5\sqrt{N} + 1.4 \lg^2 N$:
 - (a) Prepare the uniform superposition $\frac{1}{\sqrt{N}} \sum_j |j\rangle$.
 - (b) Mark every j with $T[j] < T[y]$.
 - (c) Apply the quantum exponential searching (BBHT) subroutine.
 - (d) Measure the first register; call the outcome y' . If $T[y'] < T[y]$ set $y \leftarrow y'$.
3. Return y .

The paper’s Theorem 1 states that the returned y is the index of the minimum with probability at least $\frac{1}{2}$.

2 Claims and testability

#	Claim	Type	Tested?
C1	Success prob. $\geq 1/2$ within the budget (Thm. 1).	correctness	yes
C2	Budget expression is $\lceil 22.5\sqrt{N} + 1.4 \lg^2 N \rceil$.	formula	yes
C3	BBHT expected iters = $O(\sqrt{N}/t)$ for $t \geq 1$ marked.	complexity	yes
C4	Classical baseline: $\Theta(N)$ probes.	baseline	yes

¹Also known as Grover-based amplitude amplification in the pre-BBHT era; the paper’s subroutine is what became Boyer–Brassard–Høyer–Tapp “exponential searching”.

3 Method (numbered)

1. **Environment.** Python 3.14.6 (Homebrew), NumPy 2.4.3. Pure state-vector simulation on $n = \log_2 N$ qubits; no external quantum SDK (Qiskit/Cirq/PennyLane) was used, on purpose — this ensures the replication is genuinely independent and cannot inherit any bugs from those libraries.
2. **Grover core.**
 - State: $|s\rangle = \frac{1}{\sqrt{N}} \sum_{j=0}^{N-1} |j\rangle$ — a length- N `complex128` array.
 - Oracle: phase flip of amplitudes at marked indices (`state[mask] *= -1`).
 - Diffusion: reflection about the mean amplitude (`2*state.mean() - state`), i.e. $2|s\rangle\langle s| - I$.
 - Measurement: inverse-CDF sampling from the discrete distribution $\{|\psi_j|^2\}$.
3. **Independent Grover sanity check.** For (N, k) on a 12-cell grid, ran $r^* = \text{round}(\frac{\pi}{4}\sqrt{N/k})$ Grover iterations 2000 times each and compared the empirical success rate to the closed-form $\sin^2((2r+1)\theta)$ with $\sin \theta = \sqrt{k/N}$. Worst deviation was $|\Delta| = 0.027$, mean $|\Delta| = 0.008$.
4. **BBHT subroutine.** Standard BBHT “exponential searching”: initialise $m \leftarrow 1$; on each round pick r uniformly from $\{0, \dots, \lceil m \rceil - 1\}$, run r Grover iterations on the current uniform superposition + marker, measure, and check whether the observed index is marked. On failure, $m \leftarrow \min(\lambda m, \sqrt{N})$ with $\lambda = 6/5$; otherwise return the marked index. If no items are marked (the initial threshold y already equals the minimum), the subroutine runs until the caller-supplied Grover budget is exhausted — faithful to the paper’s “infinite algorithm” idealization.
5. **Outer Dürr–Høyer loop.** Uniform-random initial y ; while total Grover iterations used $<$ budget, build `marked_mask = values < values[y]`, call BBHT with the remaining Grover budget, and adopt any returned index whose value is strictly smaller than $T[y]$.
6. **Budget.** Implemented exactly as `ceil(22.5*sqrt(N) + 1.4*log2(N)**2)` (see Table ??).
7. **Main experiment.** 5 sizes $N \in \{4, 8, 16, 32, 64\}$, 300 trials each, random permutation $T[j] = \pi(j)$ where π is a fresh Fisher–Yates shuffle of $\{0, \dots, N-1\}$. Recorded per-trial: correctness, Grover iterations used (always equals budget since we don’t early-stop), Grover iterations until y first equals the true min, and number of outer threshold updates. Seeded with 20260706; deterministic re-run.
8. **BBHT t -sweep.** $N \in \{16, 32, 64, 128\}$, $t \in \{1, 2, 4, N/8, N/4, N/2\}$, 300 trials per cell. Each trial samples a random t -subset of $\{0, \dots, N-1\}$ as the marked set, gives BBHT a Grover ceiling of $100\lceil\sqrt{N/t}\rceil + 200$ iterations (never exhausted in practice), records the total Grover iterations used until a marked index was observed.
9. **Classical baseline.** Deterministic linear scan: track running argmin, probe every element. 100 trials per $N \in \{4, \dots, 512\}$.
10. **LLM judge.** POST paper text + all evidence blobs to Argo (<http://localhost:44497/v1/chat/completions> free key `stevens`) with a strict per-claim rubric. First-preference model `argo:claude-opus-4.7` returned HTTP 502 (upstream); fell back to `argo:gpt-5.2`. Parsed JSON verdict.

4 Results

4.1 C1 — Success probability

Empirical success proportion across 300 trials per N was exactly 1.000 at every tested size. The paper’s guarantee is ≥ 0.5 ; the empirical value comfortably clears it. No failure trial was observed.

4.2 C2 — Iteration budget

N	$22.5\sqrt{N}$	$1.4\lg^2 N$	budget
4	45.00	5.60	51
8	63.64	12.60	77
16	90.00	22.40	113
32	127.28	35.00	163
64	180.00	50.40	231

Table 1: Paper budget formula versus implementation. Exact match.

4.3 C3 — BBHT subroutine scaling

Ratio $\bar{t}/\sqrt{N/t}$ where $\bar{t}$ is the empirical mean Grover iterations for BBHT to hit a marked item:

N	t	$\bar{t}$	$\sqrt{N/t}$	ratio
16	1	1.85	4.00	0.463
16	4	0.67	2.00	0.333
32	1	3.31	5.66	0.585
64	1	5.89	8.00	0.737
64	4	1.67	4.00	0.417
128	1	9.09	11.31	0.803
128	4	3.35	5.66	0.593
128	32	0.68	2.00	0.342

All 21 measured cells have ratio ≤ 0.81 . Consistent with $O(\sqrt{N/t})$.

4.4 C4 — Classical baseline

Mean probes are exactly N for all tested N ; success prob 1.0; matches the $\Theta(N)$ reference.

5 What worked / what didn’t (per claim)

C1 — worked. The full end-to-end algorithm returns the true minimum on every one of the $5 \times 300 = 1500$ trials. The paper’s $\geq 1/2$ guarantee is massively exceeded on small N .

C1 — didn’t (fully). The judge is correct that observing 300/300 does not *test tightness* of the $\geq 1/2$ bound — a stress test would need regimes where the algorithm sometimes fails. Small N (up to 64) doesn’t provide that regime. Would require reducing the iteration budget below the paper value, or scaling N several orders of magnitude.

C2 — **worked**. Formula match.

C3 — **worked**. t -sweep clean, ratios bounded.

C3 — **didn't (fully)**. Mild upward drift of the ratio at fixed t as N grows ($0.46 \rightarrow 0.80$ for $t = 1$, $N = 16 \rightarrow 128$). Consistent with the $O(\log)$ lower-order term in BBHT but I didn't quantify it against the paper's exact constant.

C4 — **worked**. Trivially and exactly.

6 Verdict + justification

PARTIAL (LLM-judge `argo:gpt-5.2`). The paper's central operational claim (success $\geq 1/2$ within the stated budget) is reproduced with margin at five tested sizes; the budget formula, the BBHT subroutine's $O(\sqrt{N/t})$ scaling, and the classical $\Theta(N)$ baseline are all directly and quantitatively confirmed. The verdict is **PARTIAL** rather than **REPLICATED** because the empirical range ($N \leq 64$) is too narrow to stress the $\geq 1/2$ bound and no confidence intervals are reported.

7 Open Questions

1. **Q1**. At what N (or reduced iteration budget) does the empirical success prob start dropping toward the $\geq 1/2$ bound? See `open_questions.json` for the concrete parameter sweep.
2. **Q2**. The BBHT ratio $\bar{v}/\sqrt{N/t}$ drifts upward at fixed t with N ; is this the $O(\log)$ lower-order term BBHT predicts or an artifact of the small- m start?
3. **Q3**. What is the empirical distribution of the number of outer-loop threshold updates, and does it match the harmonic-number H_N analysis in the paper's lemma 2?
4. **Q4**. How does the algorithm degrade under ties (non-distinct T)? Paper explicitly assumes distinct values.
5. **Q5**. Sensitivity of BBHT to λ (I used $6/5$): does another $\lambda \in (1, \sqrt{2})$ meaningfully improve total Grover-iters?

8 Reproducibility

All code and evidence in `~/Dropbox/REPLICATE-PROJECT/QC-quantum-minimum-finding-durr-hoyer-1996/`.
Re-run: `python3 work/durr_hoyer_independent.py --Ns 4 8 16 32 64 --trials 300 --out report/evid`
Total wall time on M1 Mac Mini (via CherryRd): < 2 s including all experiments.